

Registering an Embodied World to Brain Dynamics

A Cross-Modal Experiment from External 3D Capture to Biological Neural Rendering

Author: Micah Blumberg

Self Aware Networks Research Institute

<https://selfawarenetworks.com>

July 28, 2026

Abstract

Natural perception is produced while an observer moves their eyes, head, and body through a three-dimensional environment. Conventional visual-neuroscience experiments often reduce that process to two-dimensional images, fixed gaze, and isolated trials. This paper proposes a cross-modal experiment that registers external 3D scene structure, semantic and geometric object relations, gaze, head pose, body pose, action, and distributed neural measurements on a shared time base. The primary question is whether a body-conditioned recurrent 3D model predicts held-out neural state transitions and behavior better than strong 2D appearance, gaze-only, body-only, feedforward, and timing-shuffled baselines.

The protocol uses synchronized RGB-D or lidar capture, immersive extended reality, eye and body tracking, and simultaneous electroencephalography and functional near-infrared spectroscopy. A separate reduced-motion functional MRI session supplies spatial calibration rather than being treated as interchangeable with mobile recording. External computer-vision labels describe the environment; neural states are learned as uncertainty-aware components or latent trajectories and are not presumed to contain ready-made semantic labels. Viewpoint changes, occlusion, object motion, visuomotor remapping, and action-affordance changes separate retinal appearance from observer-relative structure.

The proposal develops a 2017-2022 research path in which external volumetric capture and segmentation were first joined to synchronized neural measurement and later translated into the Self-Aware Networks (SAN) concepts of Neural Rendering, Phase Wave Differentials (PWD), and Neural Array Projection Oscillation Tomography (NAPOT). The decisive SAN result would not be a compelling reconstruction image. It would be reproducible incremental prediction: receiver-relative, recurrent, multiscale neural changes must explain held-out state updates and action consequences beyond established sensory, motor, hemodynamic, and statistical alternatives. The paper defines that test, its null outcomes, and the evidence ladder required before causal perturbation is justified.

Keywords: active vision; 3D scene representation; neural rendering; NAPOT; EEG-fNIRS; virtual reality; representational similarity analysis; embodied perception; cross-modal registration; brain-computer interface

Scope and evidence boundary

Methods and preregistration paper; no participant data are reported This document specifies a proposed experiment. No participant data have been collected under this protocol. The experiment can test whether observer-relative three-dimensional scene structure helps predict distributed neural dynamics and behavior. It cannot, by decoding alone, prove that the brain contains a literal three-dimensional picture, that any one brain region is an internal screen, or that a decoded representation is conscious.

1. The Experimental Question

An ordinary visual experience exposes the problem before any specialized term is needed. Turn your head while standing in a familiar room: the image falling on the retina changes, objects disappear behind other objects, the reachable space changes with the body, and the next movement produces new sensory evidence. Yet the room is usually experienced as a stable, action-usable arrangement rather than as a succession of unrelated flat images.

The established sciences of active vision, sensorimotor control, spatial representation, predictive processing, and recurrent neural dynamics each explain parts of this achievement. The unresolved experimental problem is whether a synchronized model that explicitly joins changing three-dimensional relations, the observer's body and gaze, recurrent neural history, and returned action consequences predicts biological state transitions better than those parts considered separately.

The proposed operation is plain: measure the environment and the observer at the same time; express what is visible and actionable from that observer's current position; predict the next distributed neural and behavioral state; then change viewpoint, occlusion, or action consequences to determine which variables the prediction actually depends on. In this paper, **biological neural rendering** is the compact name for the hypothesis that a living network repeatedly performs an action-usable version of that construction. The term is earned only if the joined model survives the baselines and failure conditions below.

The central question is:

Does a synchronized, observer-relative model of external 3D structure, conditioned on gaze, head pose, body state, task, and action, predict distributed neural state transitions and subsequent behavior better than models built from two-dimensional appearance or context-free stimulus labels?

This question is deliberately narrower than the complete SAN theory. A positive result would establish that a particular class of embodied 3D model carries incremental explanatory information about measured brain dynamics. A negative result would constrain or reject that class of model. Neither result identifies a homunculus or a single location where perception is viewed.

The study separates four objects that are often conflated:

1. **The external environment:** measured surfaces, objects, spatial relations, motion, and action opportunities.
2. **The observer-relative state:** retinal input, gaze, head pose, body pose, current task, and executed action.
3. **The measured neural state:** modality-specific electrical and hemodynamic observations with different spatial and temporal transfer functions.
4. **The inferred model:** a statistical representation fitted to predict held-out neural and behavioral data.

The phrase *biological neural rendering* refers here to the hypothesis that a living network recurrently constructs action-usable relations among sensory evidence, body state, memory, and possible consequences. It does not imply a literal optical image inside the brain.

Figure 1. Proposed bridge and scale boundary. External capture and observer-relative XR rendering are established engineering operations. Synchronized external-to-neural registration is the proposed experiment. Distributed biological rendering is the later SAN / NAPOT hypothesis to be tested. Network, local-circuit, cellular, and synaptic scales are distinct; the figure does not claim that a single neuron receives every feature stream or projects to every target.

From External 3D Segmentation to Biological Neural Rendering

How volumetric media, observer-relative XR, and a proposed cross-modal BCI experiment became questions for SAN and NAPOT

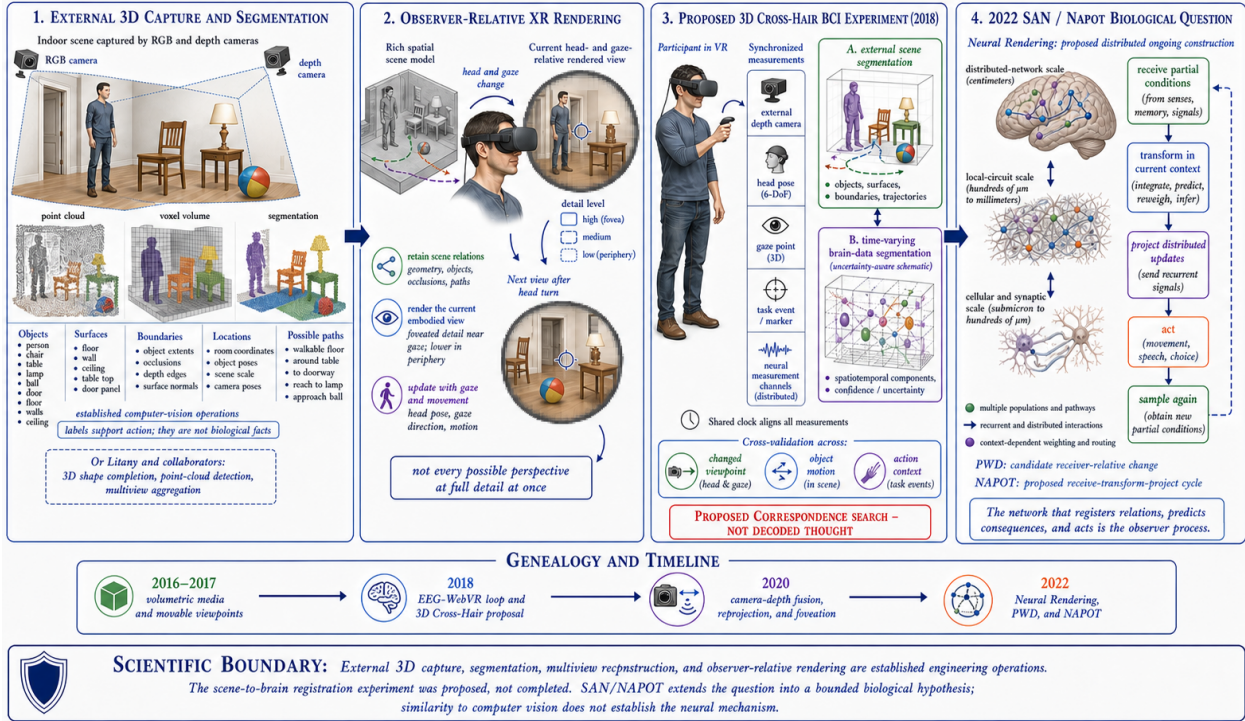


Figure 1: Figure 1. Proposed experimental bridge from external 3D capture to biological neural rendering.

2. Historical and Attribution Boundary

The experimental architecture has a dated development, but its components have different owners.

- Litany and collaborators developed methods for non-rigid 3D shape completion from partial scans. Qi, Litany, and collaborators later developed VoteNet for direct 3D object detection in point clouds.
- Tetteh, Piraud, and collaborators developed memory-efficient cross-hair filters for 3D biomedical segmentation.
- Immersive-vision researchers established methods for studying active eye and head exploration in panoramic environments.
- EEG, fNIRS, fMRI, representational similarity analysis, and neural decoding are established fields with their own methodological lineages.
- Blumberg's dated contribution is the joined application and later theoretical bridge: synchronize a segmented external 3D scene with distributed neural measurement, learn cross-modal correspondences, maintain the observer's changing perspective, and test whether a recurrent receive-transform-project account explains the resulting biological state changes.

The public sequence is:

1. **April 11, 2017:** the first Neural Lace Podcast episode described spatial computation as joining partial visual features into an object concept, using that synthesis to predict motion, and proposed placing spatial measurement around a person while also measuring the brain.
2. **May 22, 2018:** the *3D Cross-Hair Convolutional Neural Networks* article explicitly proposed correlating external 3D volumes with medical-imaging volumes and identifying neural activity associated with

captured real-world patterns.

3. **August 2018:** the public Neurohackers episode extended the question to point clouds, object segmentation, multimodal neural measurement, and the relation between viewed objects and changing brain activity.
4. **September 11, 2018:** the Jules Urbach episode joined depth capture, volumetric representation, EIT, VR, segmentation, gaze and head direction, and recurrent render-analyze-render feedback in a staged experiment.
5. **2021-2022:** the source program expanded point-cloud completion, 3D semantic segmentation, biosensors, medical imaging, and observer-relative XR, then moved the engineering problem inside the biological network.
6. **July-August 2022:** the NAPOT map and SAN Whitepaper proposed recurrent array-to-array reception, state-dependent transformation, projection, revision, and distributed rendering under the name NAPOT.

This chronology does not backdate the mature 2022 NAPOT mechanism into 2017. It preserves the earlier operational stages instead of erasing them merely because they used earlier vocabulary. Original recordings and their public platform records control meaning and provenance. Clearer 2026 retranscriptions are navigation aids whose load-bearing wording was checked against bounded audio intervals; older transcripts are discovery aids only because they contain serious recognition errors.

2.1 Companion-paper and nonduplication boundary

The companion paper *Embodied Neural Rendering in Self-Aware Networks* develops the biological theory of active recurrent vision, body maps, dendritic input arrays, Gaussian fields, frequency transitions, and the Gamma Consideration Sandwich. The present paper does not reproduce that theory. It supplies the external-to-neural measurement program: synchronized 3D scene capture, observer-relative relation graphs, versioned device clocks, leakage-free 2D-versus-3D model comparison, and the held-out ΔR_{3D}^2 endpoint.

The relationship is one-way test fan-in. H1-H5 below can succeed without establishing the Gamma Consideration Sandwich, NAPOT, or PWD. H6 can fail while an embodied recurrent 3D representation remains supported. Conversely, a theory narrative in the companion paper does not substitute for the registration, nuisance controls, and untouched confirmatory test specified here.

3. Operational Definitions

3.1 External 3D scene state

Let the external scene at time (t) be:

$$\mathcal{E}_t = \{\mathcal{P}_t, \mathcal{O}_t, \mathcal{R}_t, \mathcal{M}_t, \mathcal{A}_t\}, \quad (1)$$

where $\mathcal{P}_t$ is a calibrated point cloud or surface model, $\mathcal{O}_t$ contains object hypotheses with uncertainty, $\mathcal{R}_t$ contains spatial relations, $\mathcal{M}_t$ describes motion, and $\mathcal{A}_t$ describes candidate affordances. These are model outputs, not ground-truth properties of conscious experience.

3.2 Embodied observer state

The measured observer state is:

$$\mathcal{B}_t = [\mathbf{g}_t, \mathbf{h}_t, \mathbf{q}_t, \mathbf{u}_t, \mathbf{a}_t, \mathbf{c}_t], \quad (2)$$

where $\mathbf{g}_t$ is binocular gaze, $\mathbf{h}_t$ is head pose, $\mathbf{q}_t$ is body and hand pose, $\mathbf{u}_t$ is controller or force input, $\mathbf{a}_t$ is the executed action, and $\mathbf{c}_t$ contains task instructions and trial context.

3.3 Observer-relative scene representation

An observer-relative representation is produced by a calibrated transform:

$$\mathcal{V}_t = \Pi(\mathcal{E}_t; \mathcal{B}_t, \kappa), \quad (3)$$

where Π includes camera geometry, display timing, field of view, occlusion, visibility, and the calibration parameters κ . $\mathcal{V}_t$ may contain retinally available appearance, depth, object relations, uncertainty, and action-relevant variables. It is not assumed to equal the participant’s percept.

3.4 Latent biological state and measurement models

Let $\mathbf{z}_t$ denote an unobserved distributed biological state. Each instrument measures it through a different forward model:

$$\mathbf{y}_t^{(m)} = \mathcal{H}_m(\mathbf{z}_{t-d_m}, \eta_t^{(m)}) + \epsilon_t^{(m)}, \quad (4)$$

where m indexes EEG, fNIRS, fMRI, eye tracking, autonomic signals, or other modalities; d_m is modality-specific delay; $\eta_t^{(m)}$ contains motion, systemic physiology, and instrument state; and $\epsilon_t^{(m)}$ is residual error. EEG voltage, fNIRS hemoglobin changes, and fMRI BOLD must not be interpreted as the same variable.

3.5 Neural rendering as a predictive hypothesis

The minimal recurrent hypothesis is:

$$\hat{\mathbf{z}}_{t+\Delta} = F_\theta(\mathbf{z}_{\leq t}, \mathcal{V}_{\leq t}, \mathcal{B}_{\leq t}), \quad (5)$$

where F_θ predicts a later distributed state from prior neural, scene, and embodied history. A feedforward model is the nested special case that uses only the present sample. Evidence for recurrence requires out-of-sample improvement over that nested baseline; recurrence cannot be inferred merely because the brain contains anatomical feedback connections.

3.6 Primary endpoint

For a preregistered held-out dataset $\mathcal{D}_{test}$, define:

$$\Delta R_{3D}^2 = R^2(F_{3D+body+recurrent}) - \max_{b \in \mathcal{B}} R^2(F_b), \quad (6)$$

where $\mathcal{B}$ is the complete baseline family defined in Section 7. The primary endpoint is positive only when the confidence interval for ΔR_{3D}^2 excludes zero in an untouched confirmatory set and the gain generalizes across scenes or participants as preregistered.

3.7 Representational alignment

For each model and neural region, component, or sensor neighborhood, a representational dissimilarity matrix is:

$$D_{ij} = d(\mathbf{r}_i, \mathbf{r}_j), \quad (7)$$

and cross-modal alignment is estimated by:

$$\rho_{RSA} = \text{corr}_{\text{rank}} \left(\text{vec}(D^{\text{model}}), \text{vec}(D^{\text{neural}}) \right). \quad (8)$$

Representational similarity is supporting evidence, not identity. Shared geometry can arise from common input, task structure, motion, or analysis choices; those alternatives require explicit nuisance models and cross-validation.

4. SAN / NAPOT Formalization

4.1 Receive-transform-project motif

For a bounded neural population j , a candidate NAPOT operation can be written:

$$\mathbf{s}_{t+1}^{(j)} = \mathcal{P}_j \left[\mathcal{T}_j \left(\sum_i W_{ij} \mathcal{R}_{ij}(\mathbf{s}_t^{(i)}), \mathbf{k}_t^{(j)} \right) \right], \quad (9)$$

where $\mathcal{R}_{ij}$ denotes reception through anatomically available pathways, $\mathcal{T}_j$ is a state-dependent transformation, $\mathcal{P}_j$ is projection to downstream populations, and $\mathbf{k}_t^{(j)}$ is local context. Equation 9 is a proposed computational motif, not a claim that every cortical column implements one universal circuit.

4.2 Receiver-relative differential

A PWD-inspired event is defined relative to the expected state of a specified receiver:

$$\delta_t^{(j)} = \mathbf{f}_{\text{observed},t}^{(j)} - \hat{\mathbf{f}}_{\text{expected},t}^{(j)}, \quad (10)$$

where the feature vector may include phase, timing, rate, amplitude, duration, or multivariate population state when the instrument can measure them. An EEG phase estimate cannot establish a synapse-specific transmitter quantity, and a hemodynamic measurement cannot establish a millisecond PWD.

4.3 Multiscale constraint

The proposal predicts a structured relation across scales, not a one-to-one copy:

$$\mathbf{z}_t^{\text{macro}} = \mathcal{G} \left(\{\mathbf{z}_{k,t-\tau_k}^{\text{meso}}\}_k, \mathcal{B}_t \right), \quad (11)$$

with uncertainty in aggregation, delays, and unobserved state. A successful macro-scale model would motivate finer-scale tests. It would not prove the cellular or synaptic mechanism in Equation 9. ## 4.4 Clock transformation and causal order

Every device preserves its original time. A calibrated study time is derived by a versioned transform:

$$t_{m,k}^{study} = a_m + b_m t_{m,k}^{device} + q_m(t_{m,k}^{device}) + \varepsilon_{m,k}, \quad (12)$$

where a_m is offset, b_m is drift, q_m is any preregistered nonlinear correction, and $\varepsilon_{m,k}$ is residual alignment error. Stage 0 measures these quantities with known events. The transform and its residual distribution are frozen before neural model fitting. A channel that fails its calibration gate cannot support a confirmatory temporal-direction claim, even if a descriptive association remains visible.

Causal prediction also has a hard information boundary: a model predicting $t + \Delta$ may use external, body, and neural variables at or before t , never future gaze, future action, or future neural samples. This turns “feedback” from a diagram into a falsifiable ordering constraint.

5. Preregistered Hypotheses

H1 - Observer-relative 3D gain. A model containing depth, visibility, object relations, gaze, and body pose will predict held-out neural state better than a 2D appearance model matched for parameter count and temporal history.

H2 - Recurrent gain. A recurrent model will predict state updates after occlusion, reappearance, and viewpoint change better than a feedforward model using the same current sensory evidence.

H3 - Action dependence. Adding intended and executed action will improve prediction when identical or closely matched views permit different actions.

H4 - Cross-view generalization. A learned object or relational representation will generalize across changes in viewpoint, distance, lighting, and partial occlusion better than pixel-level similarity.

H5 - Structured update. Unexpected scene or action consequences will produce reproducible, receiver-relative changes whose timing and topology predict subsequent behavioral correction.

H6 - SAN-specific incremental fit. A preregistered PWD / NAPOT feature family will add held-out explanatory value beyond conventional rate, power, phase, event-related, hemodynamic, and predictive-error variables.

H6 is the strongest and least established hypothesis. H1-H5 can succeed while H6 fails.

6. Experimental Program

6.1 Stage 0 - Simulator and instrument validation

Before participant inference, validate the apparatus with known signals.

- Record display onset with a photodiode and controller, head, gaze, and camera events with hardware timestamps.
- Measure motion-to-photon and event-to-recording latency distributions rather than relying on nominal device specifications.
- Validate RGB-D or lidar geometry against surveyed targets.
- Inject synthetic temporal events into EEG and optical acquisition paths to test synchronization and packet loss.
- Record short-separation fNIRS channels and systemic physiology for superficial and vascular nuisance regression.

- Publish a frozen clock map and calibration report before model fitting.

6.2 Stage 1 - Behavioral XR and external 3D registration

Participants explore controlled rooms containing manipulable objects with distinct geometry and affordances. The task avoids a single iconic object example and instead uses counterbalanced sets such as a key, brush, book, cloth, lever, container, and geometric tool. Object identity, texture, location, reachability, and use are independently varied where possible.

Recorded streams include:

- stereo or RGB-D video and calibrated scene geometry;
- headset and room coordinates;
- binocular gaze and pupil measurements;
- head, torso, hand, and controller trajectories;
- contact, force, and action events;
- trial instruction, target, choice, confidence, and error;
- display-frame and acquisition timestamps.

Conditions include:

1. free active exploration;
2. passive replay of the same visual sequence;
3. viewpoint change with stable object identity;
4. object motion with stable observer pose;
5. partial and complete occlusion followed by reappearance;
6. visuomotor remapping between hand motion and scene consequence;
7. matched current images with different reachable or navigable futures;
8. object substitution that preserves low-level appearance but changes learned use.

Stage 1 establishes behavioral and engineering validity. It makes no neural claim.

6.3 Stage 2 - Reduced-motion fMRI spatial calibration

A subset of the tasks is adapted to the scanner. Active eye-gaze manipulation of panoramic video has already been demonstrated during fMRI, but head and body motion remain constrained. The fMRI session therefore supplies spatially resolved encoding and representational maps under reduced movement; it is not treated as the naturalistic experiment itself.

The session includes:

- retinotopic and category-localizer runs;
- observer-relative depth and occlusion conditions;
- action-observation and motor-planning conditions;
- matched active-choice and passive-replay sequences;
- physiological recording and motion censoring;
- subject-specific anatomical surfaces and preregistered regions plus whole-brain analyses.

6.4 Stage 3 - Mobile simultaneous EEG-fNIRS

The principal naturalistic stage combines XR, external 3D capture, eye and body tracking, EEG, and fNIRS. EEG contributes faster electrical timing at limited spatial specificity. fNIRS contributes slower cortical

hemodynamic measurements with limited depth and spatial coverage. Prior simultaneous EEG-NIRS work shows that the signals can be recorded together and can exhibit task-related correlations, while also warning that correlation does not mean both modalities measure the same neural activity.

The design uses:

- source-localized EEG only with uncertainty estimates and sensitivity analysis;
- long- and short-separation fNIRS channels;
- scalp coupling and optode-quality monitoring;
- accelerometry and inertial measurement at the cap;
- heart rate, respiration, and end-tidal or equivalent systemic measures where feasible;
- motion-matched control epochs;
- a shared event clock with empirical latency correction.

Openwater’s peer-reviewed wearable optical validation concerns cerebral hemodynamic monitoring by speckle contrast, not direct holographic recovery of neural firing. Such an optical system may become an additional read channel after independent validation for the intended task and cortical target. It must not be described as current whole-brain neural holography.

6.5 Stage 4 - Optional high-resolution modalities

MEG, intracranial ECoG, depth electrodes, or high-density electrophysiology could test finer temporal or spatial predictions in separately recruited cohorts. Intracranial data must come only from clinically indicated monitoring; the research task cannot determine electrode placement. Cross-cohort inference should use common task variables and representational geometry rather than pretending the modalities are interchangeable.

6.6 Stage 5 - Causal perturbation

Causal perturbation is deferred until the observational model passes preregistered replication. Candidate perturbations proceed from least to most invasive:

1. sensory omission or substitution;
2. visuomotor delay or remapping;
3. noninvasive stimulation with established targeting and sham controls;
4. low-intensity focused ultrasound only under a separate device, dosimetry, safety, ethics, and clinical protocol.

The read model must not automatically authorize a write target. A perturbation result is interpretable only when target engagement, sham integrity, adverse events, neural consequences, and behavior are measured independently.

7. Baseline Models

The body-conditioned recurrent 3D model is compared against all of the following:

1. **2D appearance:** pixels or image embeddings with matched temporal context.
2. **Low-level vision:** luminance, contrast, optic flow, spatial frequency, retinal eccentricity, and motion energy.
3. **3D without body:** scene geometry and object relations without gaze, head, body, or action.
4. **Body without semantics:** gaze, head, body, and action without object or relation labels.

5. **Gaze-only:** fixation, saccade, pupil, and retinal visibility variables.
6. **Task-only:** instructions, trial identity, difficulty, reaction time, and previous outcome.
7. **Feedforward:** current-state predictors with no learned history.
8. **Generic recurrent:** equally expressive temporal models without SAN-specific variables.
9. **Conventional neural features:** rate or event-related measures, spectral power, phase, connectivity, hemodynamic response, and standard prediction-error variables.
10. **Nuisance:** motion, muscle, ocular, cardiac, respiratory, superficial blood flow, drift, and device state.
11. **Timing-shuffled:** scene and neural streams shifted or permuted within prespecified constraints.
12. **Identity leakage controls:** participant, room, object, and trial identifiers withheld across splits.

The primary comparison is against the best validated baseline, not against a deliberately weak model.

8. Analysis Plan

8.1 Neural state estimation

Neural *segmentation* means the discovery of uncertainty-aware parcels, components, states, or latent trajectories from measured data. It does not mean assigning an object label directly to a brain voxel. Analyses should be repeated across reasonable parcellations and latent dimensions.

8.2 Cross-validation

Data splitting occurs before feature selection. The confirmatory hierarchy is:

1. held-out time blocks within scene;
2. held-out viewpoints;
3. held-out objects within learned relation classes;
4. held-out rooms or environments;
5. held-out participants.

Hyperparameters are selected only in nested training and validation data. The final confirmatory set is opened once.

8.3 Statistical inference

- Use participant-level effect estimates and hierarchical models rather than treating time samples as independent participants.
- Control the false discovery rate for exploratory spatial maps.
- Report uncertainty intervals, calibration, and absolute prediction performance, not only significance.
- Compare models with paired participant-level tests and permutation procedures that preserve autocorrelation.
- Determine confirmatory sample size by simulation from a preregistered pilot, including expected data loss from motion and optical-channel quality.
- Report null and negative results at every stage.

8.4 Temporal direction

Prediction must respect time. Models that use future gaze, future action, or future neural samples are descriptive smoothers, not causal predictors. Analyses should estimate whether external state precedes neural update, neural state predicts action, and returned sensory consequences update the next state, with modality-specific delays from Equation 4.

8.5 Cross-modal agreement

Cross-modal agreement is tested at the level each modality supports:

- EEG: event timing, oscillatory features, and distributed sensor or source patterns;
- fNIRS: cortical hemodynamic trajectories after systemic and superficial correction;
- fMRI: spatially resolved BOLD patterns under constrained behavior;
- behavior: gaze, choice, reaction, movement, error, and confidence;
- external model: geometry, visibility, object relations, and action consequences.

No single fusion score should conceal disagreement among modalities.

8.6 Frozen pre-data estimator smoke test

The release package includes an executable synthetic bakeoff that tests the analysis mechanics before participant recruitment. Each of forty fixed seeds generates 32 synthetic participants with four scenes and 48 steps per scene. Twenty participants train the models, six select the best baseline without access to the final set, and six remain untouched until the selected baseline is compared with the full model. The tested family includes 2D appearance, 3D without body, body without relations, a feedforward full model, a parameter-matched generic recurrent model, and the proposed 3D-body-recurrent model.

The simulator is run twice. In the alternative condition, observer-relative relational and relation-by-body terms are deliberately injected into the outcome. In the null condition, those terms are exactly zero. This is an analysis-code test, not an attempt to simulate realistic neural physiology.

Across the forty alternative runs, the mean held-out gain of the full model over the baseline selected on validation participants is 0.06925 in R^2 , the fifth percentile is 0.03400, and all forty gains are positive. Across the forty null runs, the mean is -0.00071, the 95th percentile is 0.00048, and 27.5 percent of point estimates are positive. These results show that the frozen split-selection-estimation sequence can recover the signal it was designed to recover without manufacturing a comparable mean gain when that signal is absent.

They do not estimate biological effect size, statistical power, false-positive rate under every plausible nuisance process, or the participant count required for the real study. The untouched confirmatory set in participant data may be opened only once.

8.7 Clock, motion, missingness, and latency stress test

A second pre-data program tests whether the clean bakeoff survives more hostile measurement conditions. It adds autoregressive neural history, participant heterogeneity, motion-correlated contamination, missing features, uncertain modality delay, clock jitter, and scene-wise clock drift. It repeats three explicitly synthetic scenarios at 24, 48, and 96 participants with twenty fixed seeds per cell: a reference-planning signal, a weaker signal under adverse measurement, and a true null.

In the reference scenario, the mean held-out ΔR^2_{3D} ranges from 0.04888 to 0.06595 across the three sample sizes and is positive in every run. Under adverse measurement, the mean shrinks to 0.00342-0.01353 and its diagnostic interval excludes zero in only 30-35 percent of runs. In the null, the mean remains between -0.00272 and -0.00101, with the diagnostic interval excluding zero in one of sixty cells. The result is not “more participants always rescue the study.” Misalignment, missingness, motion, heterogeneity, and delay error can attenuate the target relation until increased sample size has little practical value.

These are scenario assumptions, not pilot estimates, and the normal-approximation intervals are diagnostics rather than the final inferential method. The useful conclusion is procedural: measure clock and nuisance

distributions first, replace the planning assumptions with pilot-derived distributions, then freeze the confirmatory participant count before recruitment.

8.8 Frozen analysis contract

The accompanying preregistration contract fixes the estimand in Equation 6, the full baseline family, a participant-level 60/20/20 train-validation-test split, validation-only baseline selection, a one-time final-test opening, training-only imputation, the clock gate, identity-leakage controls, and the ordered H1-H6 confirmatory sequence. The primary fast-channel outcome is the preregistered EEG state representation; fNIRS, behavior, and reduced-motion fMRI remain key secondary outcomes with separate transfer functions rather than one opaque fusion score.

Participant-level paired effects are analyzed with a participant-cluster bootstrap or a prespecified hierarchical model. Permutations preserve participant, scene, trial block, and autocorrelation. The participant count is not guessed from these simulations: it is frozen once, after Stage 0 and a non-neural or explicitly pilot-designated stage estimate the nuisance and delay distributions, and before confirmatory recruitment. Hardware calibration, ethics approval, and pilot-derived parameters are execution gates, not missing prose in this final methods preprint.

9. Decisive Manipulations

9.1 Same appearance, different embodied relation

Create trials with nearly identical current retinal images but different body pose, reachable space, or required action. If the neural model changes only with pixels, it does not support the embodied 3D hypothesis. If action-usable state improves prediction beyond pixels and gaze, the result supports a relational representation.

9.2 Different appearance, stable object relation

Change viewpoint, illumination, or partial visibility while preserving object identity and use. Cross-view generalization tests whether the model retains structure beyond image similarity.

9.3 Occlusion and reappearance

Remove an object from view while preserving evidence about its likely location. The recurrent model should predict state during occlusion and the response to congruent or incongruent reappearance better than a feedforward model.

9.4 Visuomotor remapping

Change the mapping between participant motion and visual consequence. A body-conditioned model should track the learned mapping and its violation. The returned difference should predict subsequent correction.

9.5 Active versus replayed sampling

Replay the exact visual sequence produced by a participant while another trial removes control over gaze-contingent or action-contingent acquisition. This distinguishes sensory content from active sampling and agency-related variables.

10. Evidence Ladder

Claims advance only one rung at a time:

1. **Synchronized association:** external and neural variables covary after nuisance correction.
2. **Held-out prediction:** the model predicts untouched neural or behavioral data.
3. **Cross-view generalization:** the relation survives appearance and viewpoint change.
4. **Action-dependent update:** the model predicts effects of intended action and returned consequence.
5. **Causal sensitivity:** a controlled manipulation changes the predicted state and behavior.
6. **SAN-specific increment:** PWD / NAPOT variables add reproducible explanatory value beyond established alternatives.
7. **Mechanistic localization:** finer-scale measurements identify candidate circuits or cells consistent with the proposed operator.

A result at one rung does not license claims from a higher rung.

11. Falsifiers and Informative Null Results

The proposal is weakened or falsified in its present form if any of the following replicate:

1. 3D geometry, body state, and action provide no out-of-sample gain over 2D appearance and gaze.
2. Apparent 3D gain disappears after motion, ocular, muscular, cardiac, respiratory, and superficial hemodynamic correction.
3. The model fails across viewpoints, rooms, or participants and instead memorizes trial identity.
4. A feedforward model predicts as well as the recurrent model during occlusion and remapping.
5. Neural state does not predict subsequent action or correction after current sensory evidence is controlled.
6. PWD / NAPOT features add no reliable prediction beyond conventional neural variables.
7. The inferred latent geometry changes arbitrarily across reasonable preprocessing pipelines.
8. Causal manipulations produce no target-specific neural or behavioral effect beyond sham.

A failure of H6 does not invalidate active, embodied, 3D neural representation generally. It specifically limits the SAN extension tested here.

12. What Would Count as Support for Biological Neural Rendering?

The minimum convincing result is a preregistered replication showing that:

- observer-relative 3D and action variables improve held-out neural prediction;
- the improvement survives low-level vision, gaze, motion, systemic physiology, and task controls;
- the learned relation generalizes across viewpoint or appearance changes;
- recurrent history is necessary during occlusion or remapping;
- neural state predicts a later action or corrective update;
- the effect appears in convergent but non-identical forms across fast electrical and slower hemodynamic measurements.

Support for NAPOT would require more. The results must reveal bounded receive-transform-project relations among distributed populations, with measured delays and state-dependent transformation. Support for a PWD mechanism would require receiver-relative differential features that predict the next state or routing consequence better than established phase, power, rate, and prediction-error measures.

No reconstruction image, however recognizable, is sufficient by itself.

13. Ethics, Privacy, and Safety

This final methods preprint reports no human or animal observations. It does not claim institutional approval, participant recruitment, hardware validation, or a completed intervention. Any future participant study requires prospective ethics approval, informed consent, a public or time-stamped registration, and the execution gates specified below.

Synchronized 3D video, body motion, gaze, voice, behavior, and neural data form a highly identifying record. The protocol therefore requires:

- explicit consent for each modality and each downstream use;
- separation of identity keys from research data;
- encryption in transit and at rest;
- access logging and role-limited analysis;
- face, voice, home, and bystander protections for real-world capture;
- a prohibition on employment, insurance, policing, or covert assessment uses;
- separate consent for model sharing, raw-data sharing, and commercial use;
- participant withdrawal procedures that address trained-model derivatives;
- independent safety review for every stimulation phase.

Neural decoders are probabilistic and context-bound. Results must not be described as mind reading, thought recovery, or consciousness detection without an independently validated operational test.

14. Reproducibility Package

The study should release, subject to privacy constraints:

1. the preregistration and statistical analysis plan;
2. hardware and software versions;
3. display, sensor, and clock-calibration reports;
4. a machine-readable event schema;
5. scene geometry and semantic-label provenance;
6. preprocessing containers and parameter files;
7. train, validation, and test manifests;
8. baseline implementations;
9. model cards and failure analyses;
10. synthetic or consented benchmark data sufficient to reproduce the pipeline;
11. negative results and excluded-channel reports;
12. an immutable provenance map linking each claim to source data and code. The present release already supplies the final manuscript, the frozen analysis and preregistration contract, the event schema and example event, the original estimator smoke test, the clock / motion / missingness / latency sensitivity program, machine-readable results, source-governance receipts, validation reports, and a hash manifest. Raw source audio and machine transcripts remain in the internal audit layer and are not provider-upload files.

14.1 Minimum event schema

Each event should contain:

```
participant_pseudonym  
session_id
```

```
monotonic_timestamp
device_timestamp
clock_transform_version
scene_id
object_and_relation_state
gaze_and_head_pose
body_and_hand_pose
action_and_force
display_frame_id
neural_sample_range
physiology_sample_range
quality_flags
intervention_or_sham_state
```

14.2 Executable event contract

The final release supplies `cross-modal-event-v1.schema.json` and a valid example event. The schema requires a pseudonymous participant identifier, monotonic and device timestamps, a versioned clock transform, scene and relation state with uncertainty, gaze and body pose, action and force, display frame, modality-specific neural sample ranges, physiology range, quality flags, and intervention or sham state.

The schema deliberately records both a monotonic study clock and original device timestamps. Alignment is therefore a versioned transformation that can be audited and replaced; it is not silently baked into one timestamp. A record that satisfies the schema still does not prove that its semantic labels are correct, that the clocks are accurately calibrated, or that the corresponding neural samples contain an observer-relative representation. Those are separate validation steps.

15. Discussion

The experiment joins two research traditions without treating them as identical. External 3D computer vision reconstructs and segments measured environments for prediction and action. Systems neuroscience measures biological responses through instruments with limited transfer functions. Cross-modal registration can test whether a shared observer-relative model predicts both, but the shared model remains an inference. The design also has a specific nonduplication boundary. The companion embodied-vision paper explains active recurrent sight, body maps, dendritic receive-transform-re-express operations, and the Gamma Consideration Sandwich. This paper asks a different question: whether a synchronized external 3D and embodied state model earns incremental prediction of neural transitions under an untouched test. The companion theory supplies hypotheses; this paper supplies the registration architecture that can defeat or narrow them.

The design also corrects an EEG-only account of the earlier Neural Lace program. Fast electrical timing is important, but it is one layer of a multimodal architecture. Spatial anatomy, optical or hemodynamic state, eye and body movement, task context, and action consequences are required to interpret what EEG changes mean. Conversely, optical and hemodynamic signals cannot resolve a millisecond phase mechanism without a fast channel.

The SAN interpretation becomes scientifically useful when it makes incremental predictions. Neural Rendering predicts a continuously updated, distributed, action-usable state rather than a literal inner display. NAPOT predicts bounded receive-transform-project relations across populations rather than tomography by one generic neuron. PWD predicts receiver-relative differential events whose consequences depend on the

active local state. Each claim can fail independently.

The proposed experiment therefore does not attempt to prove consciousness by producing an attractive visualization. It asks whether embodied three-dimensional structure, recurrent state, and returned action consequences explain measurable neural transitions. That is a tractable bridge from the external capture problem first posed in the Neural Lace work to the biological mechanism later proposed in SAN.

16. Proposed Figures

1. **External 3D capture to biological neural rendering.** Four-stage architecture with explicit evidence boundaries. Draft included as Figure 1.
2. **Clock and modality transfer functions.** Display, gaze, motion, EEG, fNIRS, fMRI, and behavior on aligned but distinct timescales.
3. **Decisive 2D-versus-3D task conditions.** Same pixels / different action relation; different pixels / stable object relation; occlusion; remapping.
4. **Model bakeoff.** Nested baselines and the primary held-out ΔR_{3D}^2 endpoint.
5. **Evidence ladder.** Association through SAN-specific incremental fit and finer-scale mechanism.
6. **Receive-transform-project tests across scale.** Distributed network, local circuit, cellular, and synaptic measurements with no one-to-one identity claim. # Data and Code Availability

No participant dataset exists for this methods preprint. The release package contains the event schema, example event, frozen analysis contract, synthetic estimator code, latency-and-nuisance sensitivity code, machine-readable synthetic results, and checksums. Future participant data will be shared only to the extent allowed by consent, ethics approval, identifiability risk, and the registered data-governance plan.

Ethics Statement

No human or animal experiment is reported. Future participant stages require prospective institutional ethics approval and informed consent. Causal write channels require a separate device, dosimetry, safety, sham, adverse-event, and clinical-governance protocol.

Competing Interests

The author is the originator of Self-Aware Networks, NAPOT, PWD, and the proposed external-to-neural registration program evaluated in this manuscript. No financial competing interest is declared.

Author Contributions

Micah Blumberg: conceptualization, historical source program, proposed methodology, formalization, preregistration design, interpretation, figures, and manuscript responsibility.

Writing and Computational Assistance

AI-assisted tools were used under the author’s direction for source routing, transcription assistance, drafting support, simulation implementation, consistency checks, and document production. Original recordings and public records, not generated summaries, control source meaning and provenance. The author is responsible for the claims, interpretations, and release decision.

17. References

1. Blumberg M. [3D Cross-Hair Convolutional Neural Networks](#). *Silicon Valley Global News*. May 22, 2018.
2. Blumberg M, Urbach J. [Jules Urbach on RTX, VR capture, XR, AI rendering, and self-aware AI](#). *Silicon Valley Global News*. September 11, 2018.
3. Litany O, Bronstein A, Bronstein M, Makadia A. [Deformable Shape Completion With Graph Convolutional Autoencoders](#). *CVPR*. 2018:1886-1895.
4. Qi CR, Litany O, He K, Guibas LJ. [Deep Hough Voting for 3D Object Detection in Point Clouds](#). *ICCV*. 2019:9277-9286.
5. Tetteh G, Efremov V, Forkert ND, et al. [DeepVesselNet: Vessel Segmentation, Centerline Prediction, and Bifurcation Detection in 3-D Angiographic Volumes](#). *Front Neurosci*. 2020;14:592352.
6. Hasson U, Nir Y, Levy I, Fuhrmann G, Malach R. [Intersubject synchronization of cortical activity during natural vision](#). *Science*. 2004;303:1634-1640.
7. Nishimoto S, Vu AT, Naselaris T, Benjamini Y, Yu B, Gallant JL. [Reconstructing visual experiences from brain activity evoked by natural movies](#). *Current Biology*. 2011;21:1641-1646.
8. Kriegeskorte N, Mur M, Bandettini P. [Representational similarity analysis - connecting the branches of systems neuroscience](#). *Front Syst Neurosci*. 2008;2:4.
9. Kriegeskorte N. [Relating population-code representations between man, monkey, and computational models](#). *Front Neurosci*. 2009;3:363-373.
10. Kim HC, Jin S, Jo S, Lee JH. [A naturalistic viewing paradigm using 360-degree panoramic video clips and real-time field-of-view changes with eye-gaze tracking](#). *NeuroImage*. 2020;216:116617.
11. Haskins AJ, Mentch J, Botch TL, Robertson CE. [Active vision in immersive, 360-degree real-world environments](#). *Scientific Reports*. 2020;10:14304.
12. Bischof WF, Anderson NC, Doswell MT, Kingstone A. [Eye and head movements while encoding and recognizing panoramic scenes in virtual reality](#). *PLoS One*. 2023;18:e0282030.
13. Zama T, Shimada S. [Simultaneous measurement of electroencephalography and near-infrared spectroscopy during voluntary motor preparation](#). *Scientific Reports*. 2015;5:16438.
14. Favilla CG, Carter S, Hartl B, et al. [Validation of the Openwater wearable optical system: cerebral hemodynamic monitoring during a breath-hold maneuver](#). *Neurophotonics*. 2024;11:015008.
15. Bawiec CR, Hollender PJ, Ornellas SB, Schachtner JN, Dahill-Fuchel JF, Konecky SD, Allen JJB. [A wearable, steerable, transcranial low-intensity focused ultrasound system](#). *Journal of Ultrasound in Medicine*. 2025;44:239-261.
16. Blumberg M. [NAPOT map, immutable July 31, 2022 source](#). 2022.
17. Blumberg M. [SAN whitepaper, immutable August 5, 2022 source](#). 2022.
18. Blumberg M. [The Neural Lace Podcast, Episode 1: Democratizing Neural Lace](#). April 11, 2017.
19. Blumberg M, Rintoul J. [Neural Lace Podcast Season 2 Episode 1 Neurohackers](#). August 5, 2018.
20. Blumberg M, Urbach J. [Jules Urbach on RTX & VR, Capture & XR, AI & Rendering, and Self Aware AI](#). September 11, 2018.
21. Blumberg M. [Self Aware Networks Whitepaper, immutable GitHub Wiki revision](#). August 28, 2022.
22. Blumberg M. [Self Aware Networks Whitepaper revision history](#). 2022.
23. Blumberg M. [Embodied Neural Rendering in Self-Aware Networks: A Falsifiable Model of Vision, Sensorimotor Feedback, Body Maps, and the Gamma Consideration Sandwich](#). Zenodo; 2026. Prior public version; replacement preprint in preparation.

Appendix A. Claim Boundary Table

Claim	Current status	What this experiment can add
External systems can capture and segment 3D scenes.	Established engineering capability.	A synchronized, reproducible scene-state stream.
Eye and head movement shape active exploration.	Empirically supported.	Controlled observer-relative task variables.
EEG and fNIRS can be recorded simultaneously.	Demonstrated, with distinct transfer functions.	Naturalistic multimodal state trajectories.
External 3D state corresponds to distributed neural state.	Broadly plausible; exact proposed mapping untested.	Held-out cross-modal prediction and generalization.
Neural activity implements a recurrent action-usable rendering.	Hypothesis.	Incremental recurrent prediction under occlusion and remapping.
NAPOT's receive-transform-project motif explains the result.	SAN-specific hypothesis.	Multiscale, state-dependent directional tests.
PWD variables explain receiver-specific updates.	SAN-specific hypothesis.	Incremental prediction beyond conventional variables.
The 2017-2022 sequence is one mature NAPOT mechanism at every stage.	Not claimed; the earlier stages are operational ancestors with different vocabulary.	Passage- and audio-bound genealogy without backdating the mature term.
This paper duplicates the embodied-vision companion paper.	Rejected by scope audit.	This paper tests synchronized external-to-neural registration; the companion develops the detailed biological theory.
The decoded state is conscious.	Not established by decoding.	Requires a separate operational consciousness protocol.

Appendix B. Execution Gates After Publication

The final preprint completes the work that can be completed before hardware validation and participant recruitment: an exact event contract, two synthetic analysis checks, a frozen analysis and preregistration contract, source-faithful genealogy, a Whitepaper image-to-claim map, the nonduplication boundary with *Embodied Neural Rendering in Self-Aware Networks*, an ethics boundary, and a reproducible release manifest. The following are prospective experiment-execution gates, not unfinished manuscript tasks:

1. Add an immutable execution addendum after Stage 0 that freezes numerical clock, geometry, gaze, packet-loss, EEG, optical-coupling, motion, and missingness thresholds.
2. Extend the event schema into device-specific records and complete the hardware latency bench.
3. Replace the scenario assumptions with pilot-derived nuisance and delay distributions, then freeze the confirmatory participant count before recruitment.
4. Select a medically and ergonomically reviewed XR/fNIRS / EEG arrangement.
5. Run the non-neural Stage 1 pilot to validate geometry, gaze, body tracking, task contrasts, privacy handling, and data loss.
6. Obtain prospective institutional ethics approval and register the observational protocol before collecting confirmatory participant data.
7. Defer ultrasound and every other write channel until the observational model replicates and a separate device, dosimetry, safety, sham, adverse-event, and clinical protocol is approved.

None of these future gates changes the status of this document as a complete methods and preregistration preprint. They determine whether and how the proposed experiment may be executed.